

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.94%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.21%	Share of actual attacks successfully flagged
F1-Score	98.56%	Balanced measure combining Precision and Recall
Accuracy	99.05%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.03%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.79%	Actual attacks that went undetected
True Positive Rate (TPR)	97.21%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.97%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4782	FN 137
Actual: Normal	FP 3	TN 9800

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.94%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.97%	Balanced measure combining Precision and Recall
Accuracy	99.94%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	100.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	0.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	4782	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.94%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.21%	Share of actual attacks successfully flagged
F1-Score	98.56%	Balanced measure combining Precision and Recall
Accuracy	99.05%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.03%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.79%	Actual attacks that went undetected
True Positive Rate (TPR)	97.21%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.97%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4782	FN 137
Actual: Normal	FP 3	TN 9800

2d. Response Latency

Metric	Value	Description
Detection Time	522.2 ms	Average time taken to detect an attack
Mitigation Response Time	10.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.01%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	7.49%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	4.89%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.23	5	4	2026-09-26 23:27:33	2026-09-26 23:30:13
10.0.0.25	5	3	2026-09-26 23:27:38	2026-09-26 23:30:30
10.0.0.24	4	3	2026-09-26 23:27:34	2026-09-26 23:30:17
10.0.0.9	2	2	2026-09-26 23:26:35	2026-09-26 23:27:46

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 23:26:35	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 23:27:05	10.0.0.9	DDoS	UDP Flood	92.4%	High	Released
2026-09-26 23:27:33	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 23:27:34	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 23:27:35	10.0.0.9	DDoS	UDP Flood	92.4%	High	Quarantined
2026-09-26 23:27:38	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 23:27:44	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (30s)
2026-09-26 23:27:44	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-26 23:27:45	10.0.0.24	DDoS	ICMP Flood	91.3%	High	Time Ban (30s)
2026-09-26 23:27:45	10.0.0.24	DDoS	ICMP Flood	91.3%	Critical	Time Ban
2026-09-26 23:27:49	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (30s)
2026-09-26 23:27:49	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-26 23:28:14	10.0.0.23	DDoS	ICMP Flood	94.6%	High	Released
2026-09-26 23:28:19	10.0.0.25	DDoS	ICMP Flood	92.5%	High	Released
2026-09-26 23:28:25	10.0.0.23	DDoS	ICMP Flood	94.6%	Critical	Time Ban (1m)
2026-09-26 23:28:30	10.0.0.25	DDoS	ICMP Flood	92.5%	Critical	Time Ban (1m)
2026-09-26 23:28:45	10.0.0.24	DDoS	ICMP Flood	92.1%	High	Time Ban (1m)
2026-09-26 23:29:26	10.0.0.23	DDoS	ICMP Flood	94.6%	Critical	Time Ban (2m)
2026-09-26 23:30:13	10.0.0.23	DDoS	ICMP Flood	90.5%	Critical	Time Ban (5m)
2026-09-26 23:30:17	10.0.0.24	DDoS	ICMP Flood	90.6%	Critical	Time Ban (2m)
2026-09-26 23:30:30	10.0.0.25	DDoS	ICMP Flood	90.1%	Critical	Time Ban (5m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.25	ICMP Flood	0.6454	90.1%	Critical	Phase 2	1	11s	2026-09-26 23:30:30	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6429	90.6%	Critical	Phase 2	1	10s	2026-09-26 23:30:17	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6454	90.5%	Critical	Phase 2	1	10s	2026-09-26 23:30:13	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6444	94.4%	Critical	Phase 2	1	1m 11s	2026-09-26 23:29:31	Ban Expired
10.0.0.23	ICMP Flood	0.6448	94.6%	Critical	Phase 2	4	1m 12s	2026-09-26 23:29:26	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6424	92.1%	High	Phase 2	3	1m 11s	2026-09-26 23:28:45	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.25	ICMP Flood	0.6461	92.5%	Critical	Phase 2	3	10s	2026-09-26 23:28:30	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6458	94.6%	Critical	Phase 2	3	11s	2026-09-26 23:28:25	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6138	92.5%	High	Phase 2	1	41s	2026-09-26 23:28:19	Ban Expired
10.0.0.24	ICMP Flood	0.6138	91.7%	High	Phase 2	2	41s	2026-09-26 23:28:15	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6337	94.6%	High	Phase 2	1	41s	2026-09-26 23:28:14	Ban Expired
10.0.0.25	ICMP Flood	0.6456	91.2%	High	Phase 2	1	11s	2026-09-26 23:27:49	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6114	92.4%	High	Phase 1	2	11s	2026-09-26 23:27:46	Attack Stopped
10.0.0.24	ICMP Flood	0.6464	91.3%	High	Phase 2	1	11s	2026-09-26 23:27:45	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6454	90.8%	High	Phase 2	1	11s	2026-09-26 23:27:44	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6182	92.4%	High	Phase 2	1	30s	2026-09-26 23:27:05	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE